

The Development of Indonesian Capitalism

Zhihu column · 2026-08-08

I remember that last year, when large-scale protests broke out in Indonesia, Nepal, and other countries, someone published an article lamenting the fate of the people of "semi-colonial and semi-feudal countries" [1]. At the time I felt somewhat puzzled. To call Nepal a semi-colonial, semi-feudal country may have some justification — after all, its capital accumulation and level of industrialization are both quite low. But to call Indonesia a semi-colonial and semi-feudal country seems far less certain. For a semi-colonial, semi-feudal society is generally an agrarian country that has not completed industrialization, dominated by agriculture — yet Indonesia, after decades of development, can hardly be called an agrarian country anymore. Moreover, from my own impression traveling in Indonesia, the country's level of development is basically comparable to that of an average Chinese province in the 2000s — hard to connect with the picture of a semi-colonial, semi-feudal society struggling outside the gates of industrialization. This article explores the development of Indonesian capitalism at every level.

I. Indonesia's Level of Economic Development

1. Main Physical Indicators of Industry

As the figure below shows, apart from steel and cement, which lag behind — at about the level of China in the early 1990s — Indonesia's output of major industrial products has basically reached, or come close to, China's level of the early 2000s.

国家(年份)	总人口 (百万)	百万人钢产量 (吨)	百万人煤炭产 量(吨)	人均发电量(千 瓦时)	百万人炼油 量(吨)	百万人汽车产 量	百万人PVC产 量(吨)	百万人水泥 产量(吨)
印尼2025	288	65973	283681	1584	177778	3985	1737	224653
中国1990	1134	58509	95239	551	94357	449	693	184930
中国1995	1218	78293	106569	770	115764	1191	1129	366175
中国2000	1265	100582	79052	993	166799	1627	1395	454937
中国2005	1304	272846	167945	1782	219326	4372	4979	812884
中国2010	1340	467646	241792	2944	315672	13627	8433	1394030

Indonesia is now making great efforts, through attracting foreign investment, to fill the gaps in heavy industry.

In 2025, Indonesia's stainless steel capacity exceeded 5 million tonnes, making it the world's second-largest producer; per capita it has already surpassed China's 2025 level. The stainless steel producers

in Indonesia are mainly Chinese and Indian enterprises. Korean firms also plan to join hands with Chinese enterprises to build a 2-million-tonne stainless steel plant in Indonesia ^[2]. In the coming years, Indonesian steel capacity will keep expanding, reaching 30–35 million tonnes by 2030, which per capita exceeds China's 2000 level.

In refining, projects under way in Indonesia include two state-owned or joint-venture refineries ^[3] ^[4] and a new refinery a Chinese firm plans to build ^[5], adding roughly 25 million tonnes of annual refining capacity — per capita, above China's 2005 level.

In electricity, Indonesia will add nearly 70 GW of generating capacity over the next decade, more than half of it green power ^[6]. If all this capacity materializes, annual per-capita generation will rise by about 900 kWh, to about 2,000 kWh — equivalent to China's level in 2008.

In short, on the evidence of physical industrial production, Indonesia has essentially reached industrial-country level; but its industrialization is not yet complete. It is still in a process of rapid industrial development, and will need roughly fifteen more years to reach about China's 2012 level and essentially complete industrialization.

2. The Level of Infrastructure

Indonesia is an archipelago whose population is scattered across tens of thousands of islands; the conditions for large-scale infrastructure development are not as good as China's. Still, Indonesia's population is concentrated on Java and Sumatra, which together account for nearly 60 percent and over 20 percent of residents — more than 80 percent of the country. Java, with an area of nearly 140,000 square kilometers and 140 million people, has a high level of economy and can be compared with Guangdong Province (nearly 180,000 square kilometers, about 100 million people). Sumatra is larger (470,000 square kilometers) but economically less developed; it can be compared with Sichuan Province (480,000 square kilometers).

(1) Java versus Guangdong

In 2025, Java's road network totaled 120,000 kilometers — equivalent to Guangdong's level in 2005 (115,337 km). Java's expressways reached 1,782 km, comparable to Guangdong's 2002 level (1,885 km).

In 2025, Java's railways totaled nearly 4,700 km, reaching Guangdong's 2015 level (4,021 km). Java's first high-speed rail, the Jakarta–Bandung HSR, is already open, 142 km long. Guangdong's first high-speed line was the Wuhan–Guangzhou HSR, opened at the end of 2009.

In 2025, Java has one metro line in operation, 15.7 km; and two light-rail lines (including one linking Jakarta's satellite cities), about 50 km in all. In 2004, Guangdong had nearly 59 km of metro in

operation and no light rail.

In 2025, Java's largest port, Tanjung Priok in Jakarta, will handle about 8 million TEU — equivalent to Shenzhen Port's level in 2002 (7.61 million TEU).

On the whole, Java's infrastructure has reached the level of China's most developed provinces in the mid-2000s.

(2) Sumatra versus Sichuan

In 2025, Sumatra's roads totaled about 93,000 km — equivalent to Sichuan's level in 2000 (91,000 km). Sumatra's only high-standard trunk road network is the Trans-Sumatra Toll Road (JTTS); by the end of 2025 about 1,235 km had opened, comparable to Sichuan's 2003 level (1,501 km).

In 2025, Sumatra's railways totaled only about 1,340 km — equivalent to Sichuan's level in 1965 (1,530 km). Sumatra has no high-speed rail; Sichuan's first HSR line opened in 2010.

In 2025, Sumatra has only one urban light-rail line (the Palembang LRT), 23.4 km. Sichuan opened its first metro line in 2010, 18.5 km long. In rail transit, then, Sumatra is roughly at Sichuan's level around 2010.

On the whole, Sumatra's infrastructure has reached the level of China's western provinces in the 2000s, but its railway construction lags badly.

For Java and Sumatra — the home of 80 percent of Indonesia's population — the infrastructure level is basically equivalent to China's 2000s. The Indonesian government is now pushing road, rail, and port construction hard; several key projects on Java and Sumatra are under way:

- Trans-Sumatra Toll Road: planned total length about 2,818 km (sections of phases 1 and 2 are already open; work is racing on the remaining ~1,000 km of critical mainline gaps and spurs), scheduled for full completion in 2031.
- Trans-Sumatra Railway main network: over a thousand kilometers of new track to integrate the existing ~1,200 km of scattered, provincial freight lines into one network. The first phase, Banda Aceh–Besitang, 478 km, is planned for completion before 2030.
- Jakarta–Bandung HSR extension (Jakarta–Bandung–Yogyakarta–Surabaya): the extension is about 570 km (building on the 142 km already open, extending east across Java to the second city, Surabaya); because of the huge investment, completion is expected only in the 2030s.
- Jakarta MRT East–West Line (Phase 3): about 84.1 km (crossing Banten, the Jakarta special capital region, and West Java), planned to open in 2031.

- Jakarta MRT North–South Line Phase 2 (extension): about 11.8 km (2A section 5.8 km and 2B section 6.0 km, with tunnel boring under way), planned to open in 2029.
- Patimban Deep-Sea Port (Phase 2 and 3 expansion): planned 7.5 million TEU capacity (built jointly by Indonesia and Japan to relieve the old Jakarta port area; final completion will bring it to that container capacity), scheduled for 2027.
- New Tanjung Priok Port (Kalibaru port area expansion): 15–18 million TEU (on the current limit of about 8 million TEU, building new automated container terminals CT2, CT3, etc., to double total capacity), scheduled for 2030.

3. Monetary Indicators

In 2025, Indonesia's manufacturing gross output was \$276.5 billion; converted at 0.61 to 2005 dollars, that is \$168.665 billion in 2005 dollars. In 2005, China's manufacturing GDP was \$736 billion. Indonesia's population in 2025 is 288 million; China's in 2005 was 1.3 billion. Adjusted by population ratio, Indonesia's manufacturing output corresponds to \$761.3 billion in China-2005 terms — on par with China's 2005 level.

Indonesia's total merchandise exports in 2025 were \$282.91 billion; converted at 0.61, that is \$172.5 billion in 2005 dollars. China's merchandise exports in 2005 were \$762 billion. By population ratio, Indonesia's 2025 exports correspond to China-2005 exports of \$778.6 billion — essentially on par with China's 2005 level.

Likewise, Indonesia's 2025 per-capita GDP of \$5,083, converted at 0.67 to 2008 dollars, is \$3,405. China's 2008 per-capita GDP was about \$3,300 at then-current exchange rates. Indonesia's 2025 per-capita GDP is therefore essentially equivalent to China's 2008 level.

On the whole, Indonesia's monetary indicators broadly match its physical industrial indicators and infrastructure level — roughly equivalent to China's mid-2000s.

II. Indonesia's Industrial and Class Structure

1. Is Indonesia an Industrial Country?

In 1990, agriculture fell below 20 percent of Indonesia's GDP ^[7]; in China, agriculture's share did not fall below 20 percent until 1993. Today agriculture is below 15 percent of Indonesia's GDP, comparable to China in the early 2000s.

Agriculture, forestry, and fishing, value added (% of GDP) - Indonesia, China

Country official statistics, National Statistical Organizations and/or Central Banks; National Accounts data files, Organisation for Economic Co-operation and Development (OECD); Staff estimates, World Bank (WB)

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Agriculture, forestry, and fishing, value added (% of GDP) - Indonesia, China

As for manufacturing, its share of Indonesian GDP was once comparable to China's in the early 2000s, above 30 percent [8]. After China joined the WTO, its competition hammered the industry of the whole of Southeast Asia, driving manufacturing's GDP share steadily down. In recent years, however, with large-scale relocation of Chinese industry outward, Indonesia's manufacturing share has stabilized and picked up slightly. But since the 1990s, manufacturing's share of GDP has stayed above agriculture's; counting construction, industry's share of GDP is far above agriculture's.

Manufacturing, value added (% of GDP) - Indonesia, China

Country official statistics, National Statistical Offices (NSOs); National Accounts data files, Central Banks; Staff estimates, World Bank (WB)

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Manufacturing, value added (% of GDP) - Indonesia, China

By the standard China itself historically used for an industrial country — industrial output exceeding agricultural output — Indonesia has been an industrial country since the 1990s. By the "Chenery criteria," with agriculture below 20 percent of GDP and approaching 10 percent, Indonesia is in the latter half of the middle stage of industrialization, not far from completing it (agriculture below 10 percent of GDP).

2. The Share of the Proletariat: Indonesia in 2025 versus Tsarist Russia in 1913

In Tsarist Russia in 1913 ^[9], the total population was 170 million ^[10], of whom about 52 percent, or 88 million, were of working age. All non-agricultural proletarians numbered 15 million — 17 percent of the working-age population. Counting only manufacturing and construction, there were about 7

million — 8 percent of the working-age population. Adding the 6.5 million wage laborers in agriculture ^[11], total hired workers came to 21.5 million — 24.4 percent of all employed persons.

Mining	645,000
Construction	1,600,000
Small industries	3,706,000
Commerce	865,000
Household	3,000,000
Menial laborers	2,500,000
State-owned armaments, shipbuilding, etc.	1,500,000
Approximate total	15,000,000

Table 1: Hired Non-Agricultural Manual Labor, Usually Omitted Categories (early 1914)⁵

Wage and salaried workers in Indonesia are close to 50 percent of total employment ^[12], roughly China's level in 1999 — far above pre-war Tsarist Russia, and comparable to France around 1900.

Wage and salaried workers, total (% of total employment) (modeled ILO estimate) - Indonesia, China

ILO Modelled Estimates database (ILOEST), International Labour Organization (ILO), uri: ilostat ilo.org/data/bulk, publisher: ILOSTAT, type: external database, date accessed: January 17, 2026

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Wage and salaried workers, total (% of total employment) (modeled ILO estimate) - Indonesia, China

Industrial employment in Indonesia has reached about 23 percent of total employment, comparable to China's level between 1995 and 2005 ^[13].

Employment in industry (% of total employment) (modeled ILO estimate) - Indonesia, China

ILO Modelled Estimates database (ILOEST), International Labour Organization (ILO), uri: ilostat.ilo.org/data/bulk, publisher: ILOSTAT, type: external database, date accessed: January 17, 2026

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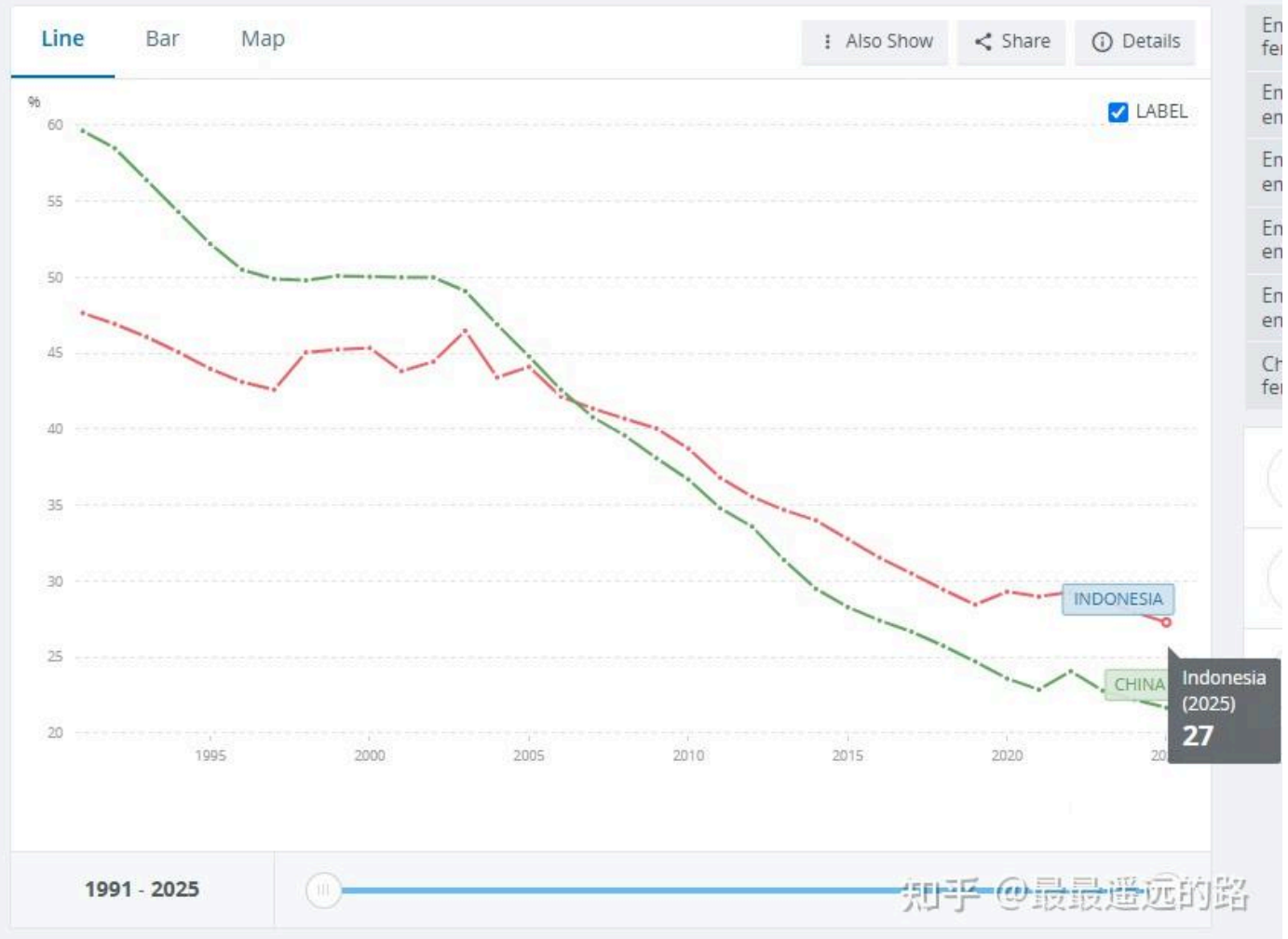
Employment in industry (% of total employment) (modeled ILO estimate) - Indonesia, China

Agricultural employment in Indonesia is now only 27 percent ^[14], equivalent to China's 2017 level. In Tsarist Russia in 1913, by contrast, agricultural employment was about 75 percent — far above even Indonesia's pre-1990s level.

Employment in agriculture (% of total employment) (modeled ILO estimate) - Indonesia, China

ILO Modelled Estimates database (ILOEST), International Labour Organization (ILO), uri: ilostat.ilo.org/data/bulk, publisher: ILOSTAT, type: external database, date accessed: January 17, 2026

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Employment in agriculture (% of total employment) (modeled ILO estimate) - Indonesia, China

In the share of the proletariat, the industrial proletariat, and the share of agricultural employment, Indonesia in 2025 is far more thoroughly capitalized than Tsarist Russia in 1913, and its overall level is closer to France and Germany around 1900. Given the actual class structure, it would be a grave distortion of reality to regard Indonesia as a country that has not yet completed its capitalist transformation.

3. How Much Feudal Residue Is Left in Indonesian Agriculture?

According to survey data provided by Indonesian scholars, most Indonesian peasants are owner-cultivators ^[15]; about 30 percent own no land and must rent it.

Table 5.2 Land management and sharecropping.

Region	Own	Lease	Sharecropping	Ratio
Sumatra	14,452 (23.39%)	5,826 (36.90%)	3,101 (33.16%)	2
Java	24,708 (39.99%)	4,976 (31.52%)	3,319 (35.49%)	3
Kalimantan	6,646 (10.76%)	1,243 (7.87%)	829 (8.87%)	4
Sulawesi	7,535 (12.20%)	1,414 (8.96%)	676 (7.23%)	4
Bali & Nusa Tenggara	7,064 (11.43%)	2,222 (14.07%)	1,270 (13.58%)	3
Maluku	747 (1.21%)	72 (0.46%)	57 (0.61%)	6
Papua	632 (1.02%)	35 (0.22%)	99 (10.6%)	5
Indonesia	61,784 (70.75%)	45,788 (18.08%)	9,351 (10.71%)	3

Source: Statistics Indonesia (BPS), 2014.

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By comparison, in Japan in 1941, 30 percent of peasants had no land at all, and another 40 percent had to rent at least part of their land ^[16]. Moreover, peasants then were more than 40 percent of all employed persons in Japan — far above Indonesia today. Indonesia's feudal residue is thus no greater than Japan's before the land reform, and calling such a society "semi-feudal" is a stretch.

第2表 農地改革前後における自小作別農家戸数の変化
(都府県), 1941-50年

年	農家戸数(千戸)				
	合計 ^a	自作 ^b	自小作	小自作	小作
1941	5,203 (100)	1,589 (31)	1,095 (21)	1,080 (21)	1,439 (28)
1947	5,700 (100)	2,068 (36)	1,157 (20)	985 (17)	1,489 (26)
1950	5,895 (100)	3,652 (62)	1,569 (27)	403 (7)	271 (5)

出典：農林省(1952, pp. 55-60).
注：^a例外規定農家(土地を耕作しない)を除く
^b耕作地主を含む。

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As for age structure, young people are fewer than the middle-aged and old, and over the last decade the overall trend remains one of young people steadily leaving agriculture for urban wage labor ^[17]. This shows that the pool of labor available to land-based exploitation keeps shrinking; whatever feudal residue remains is in the process of disappearing.

Table 3. Number of individual agricultural holdings by age group in Indonesia, ST 2013 and ST 2023 (household)

Age group (years)	ST 2013		ST 2023		Growth (%)
	HH	%	HH	%	
<15	3,297	0.01	728	0.00	-77.92
15-24	229,943	0.88	363,327	1.24	58.01
25-34	3,129,644	11.97	3,005,033	10.24	-3.98
35-44	6,885,100	26.34	6,477,990	22.08	-5.91
45-54	7,325,544	28.03	7,947,405	27.09	8.49
55-64	5,229,903	20.01	6,808,739	23.20	30.19
≥65	3,332,038	12.75	4,738,980	16.15	42.22
Total	26,135,469	100	29,342,202	100	12.27

Source: BPS, 2013 and BPS, 2023

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III. Is Indonesia a Semi-Colonial Country?

A "semi-colonial" country is one that is formally independent but whose sovereignty is in reality incomplete. The most typical example is China from 1840 to 1949. What rights of sovereignty did China lose? To speak only of the economic, the two main ones were: loss of tariff autonomy, and loss of consular jurisdiction in the concessions. On the one hand, foreign firms could pour goods into the Chinese market paying only very low tariffs. On the other hand, foreign firms in China that operated in the concessions did not have to pay taxes to the Chinese government. Both matters not only sapped the competitiveness of Chinese firms but weakened the state's fiscal capacity, to the detriment of a latecomer's industrialization. Before and during the Second World War, China gradually recovered its tariff rights and concessions, but the economic chaos of the civil war made it impossible to exercise these rights effectively.

By this standard, can Indonesia be counted a semi-colonial country?

First, on tariff rights: Indonesia has frequently adjusted tariffs in recent years ^[18] ^[19] to protect domestic industry or to encourage foreign production on its soil. We have not seen any imperialist power threatening Indonesia with force, economic sanctions, or any other means to lower its tariffs. Of course, tariff wars among capitalist countries are perfectly normal and cannot by themselves mark a country as semi-colonial.

To induce some firms to produce locally, the Indonesian government has gone so far as to ban their products ^[20]. Even Apple, powerful as it is, did not dare to have its home country send warships to blockade the ports in defense of "free trade."

为解除“封杀”，苹果妥协了！据称将在印尼投资1000万美元

原创

环球市场情报

2024-11-05 14:12 星期二 财联社 黄君芝 责编 林琦

- ①苹果公司计划在印度尼西亚投资近1000万美元，以扩大其产品生产；
- ②印尼政府此前宣布禁止在该国销售最新款iPhone 16，原因是苹果未能履行其在印尼的投资承诺。

财联社11月5日讯（编辑 黄君芝）据知情人士透露，苹果公司（Apple Inc.）计划在印度尼西亚投资近1000万美元，以扩大其产品生产，进而寻求解除印尼对其最新款iPhone的销售禁令。

这名知情人士说，该计划将涉及苹果与其供应商合作在雅加达东南部的万隆投资一家工厂。他表示，该工厂将生产苹果设备的配件和零部件等产品。印尼商务部正在审议这一提议，预计很快就会做出决定。这一提议尚未最终确定，可能会有变化。

很显然，对于印尼的“封杀”，苹果这回是妥协了。**印尼政府此前宣布禁止在该国销售最新款iPhone 16，原因是苹果未能履行其在印尼的投资承诺。**印尼政府要求苹果在印尼实现40%的国产化率，但苹果公司在印尼的工厂无法满足这一要求。

Second, on taxation: every foreign enterprise entering Indonesia must pay taxes; no one has heard of any company doing business in Indonesia without paying taxes.

Finally, to maximize the use of its own resources, the Indonesian government has begun to implement a "state capitalism" model, mandating that exports of major resource products be channeled through state enterprises ^[21]. At the same time, the government requires foreign mining companies to dilute their equity year by year, until foreign stakes ultimately do not exceed 49 percent ^[22]. Faced with such "high-handed" clauses, no country — including China — has yet dared to send warships to defend "free trade" and "private property."

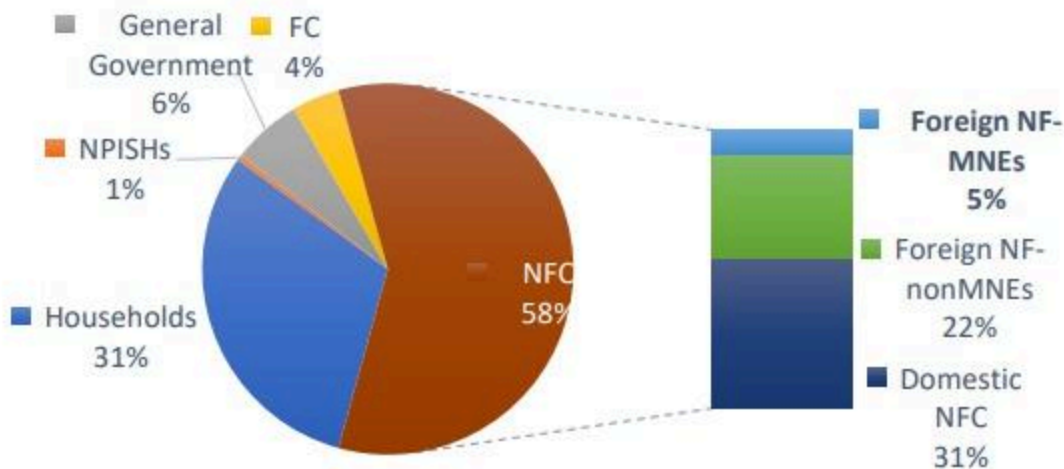
On these grounds we can judge that Indonesia is definitely not a semi-colonial country. In other words, a semi-colonial country could not do the things the Indonesian government does. Indonesia is, in fact, universally regarded internationally as a "middle power." If such a country could be labeled "semi-colonial," then Japan, Canada, and the rest could all be counted semi-colonial, and few sovereign states would remain in the world. The concept of semi-colonialism would become meaningless.

Of course, some will say that foreign capital accounts for a high share in Indonesia. But in reality, like other Third World countries, Indonesia long operated a mixed economy with a very high share of state-owned enterprises. Even as late as 2010, "40 percent of Indonesia's GDP was produced by state enterprises; in 2010, 141 state enterprises paid up taxes and profits totaling 120 trillion rupiah (about \$13.5 billion), 18 percent of government revenue" ^[23]. With large-scale privatization of state

enterprises, this share has fallen a great deal ^[24]. State banks still hold more than 40 percent of banking financial assets and dominate the sector ^[25].

In 2019, foreign enterprises contributed 26 percent of Indonesia's GDP; the remaining 74 percent was contributed by domestic enterprises ^[26]. By comparison, in China "in 2010, the total assets of industrial enterprises above a designated size were 59.29 trillion yuan, of which state industrial enterprises held 41.8 percent, private industrial enterprises 33.1 percent, and foreign enterprises 25.1 percent." If foreign capital's returns in Indonesia are comparable to domestic capital's, foreign capital's share of capital is also around 26 percent — a little higher at most, about the same as China around 2010. There is no situation in which foreign capital controls the whole of Indonesia, reducing it to semi-colonial status.

Share of foreign NF MNEs GVA to GDP 2019



Foreign-controlled NFC contributed 26% to GDP of which foreign NF MNEs share is 5%

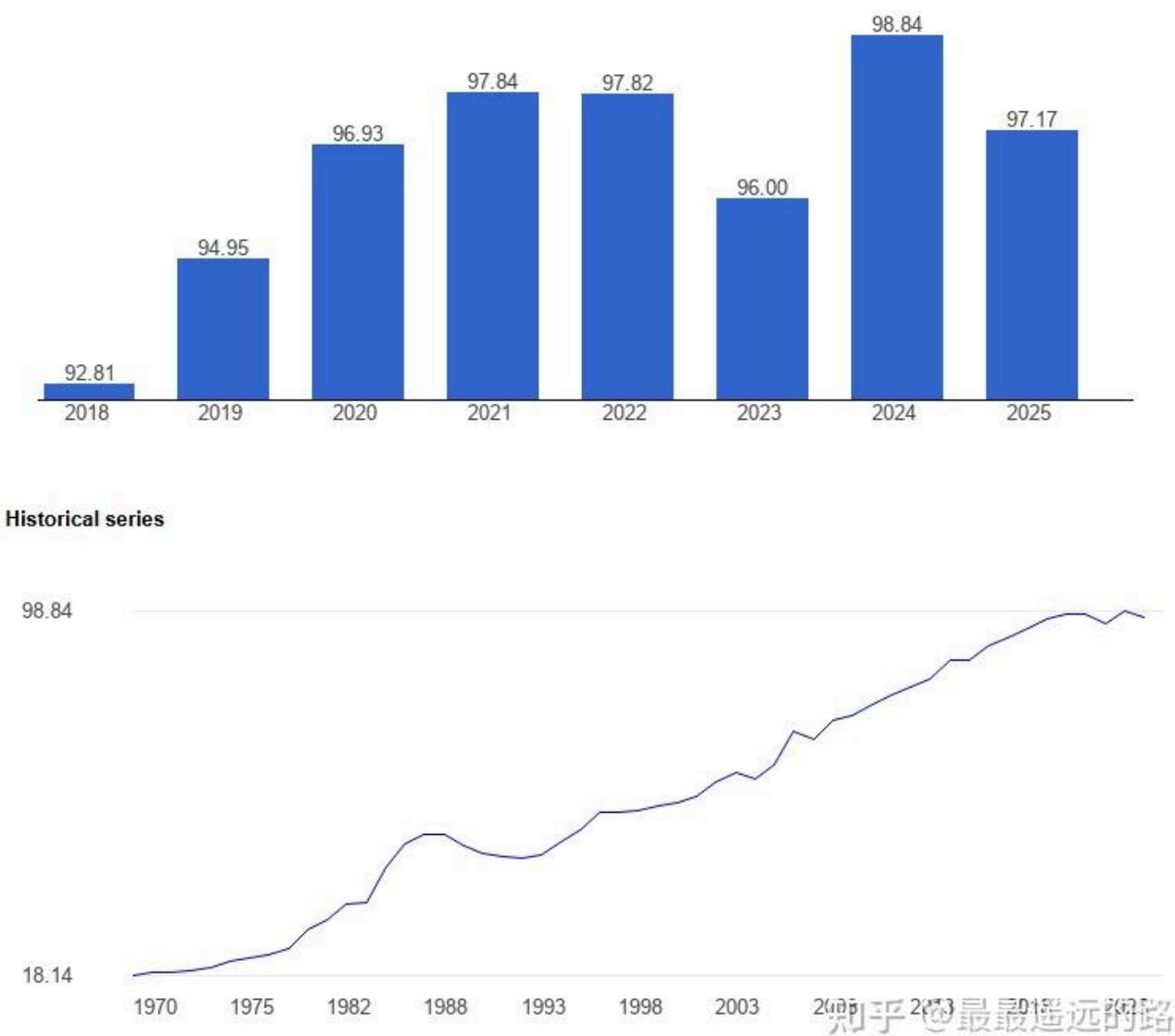
In fact, in the age of globalization, the fact that foreign capital is willing to enter a country only shows that the country is socially stable and its economic prospects good. Conversely, a country with essentially no foreign investment is quite likely one that is isolated from the world market. If Indonesia is "semi-colonial," then China in the 2000s could equally be called semi-colonial. Older

readers may remember that some grand old men of the Left used to preach this very theory, fearfully worrying that the country would become an economic colony of the United States and Japan. A dozen years later, when the trade war broke out and foreign capital was really about to leave, the same people were saying America was waging strategic encirclement.

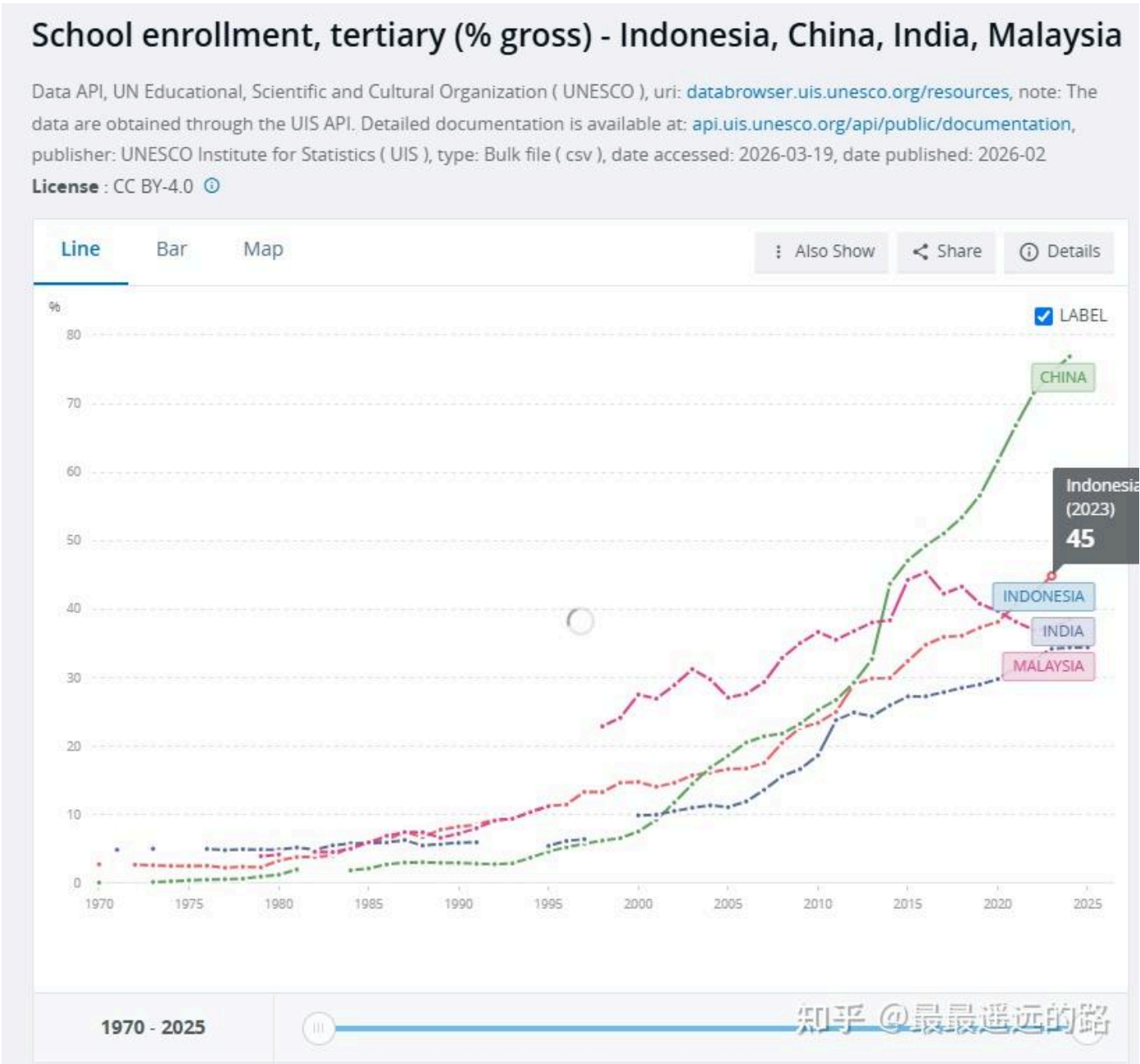
IV. Prospects for Development

1. Labor

Indonesia's gross secondary school enrollment has reached 97 percent ^[27] — basic education is essentially universal, fitting Indonesia's current stage of developing large-scale mid- and low-end manufacturing.



Indonesia's gross tertiary enrollment reached 45 percent in 2013 — higher than India, even higher than Malaysia, equivalent to China's 2014 level, within a decade of China [28]. Compared with its overall industrial level, Indonesia's higher education is developing ahead of the curve.



School enrollment, tertiary (% gross) - Indonesia, China, India, Malaysia

In 2020, Indonesia graduated 300,000 students in science and engineering, ranking fifth in the world [29]. As university enrollment keeps rising, the number of STEM graduates will grow further, eventually reaching about 500,000 a year. This provides the human base for Indonesia to develop indigenous mid- and high-end industries.

Top Countries by Number of STEM Graduates

Graduates in Science, Technology, Engineering, and Mathematics in 2020



Figure 1: Top Countries by Number of STEM Graduates in 2020. (Original graph created by [Niall McCarthy at Statista](#))

2. R&D Spending

For a long time, Indonesia's R&D spending has been the lowest among the major economies — lower even than countries at a lower overall level of development, such as India ^[30].

R&D Spending in G20 nations as % of GDP		
Rank	Country	R&D spending
1	South Korea	4.93%
2	United States	3.46%
3	Japan	3.30%
4	Germany	3.14%
5	United Kingdom	2.91%
6	China	2.43%
7	France	2.22%
8	Australia	1.83%
9	Canada	1.55%
10	Italy	1.45%
11	Turkey	1.40%
12	Brazil	1.15%
13	Russia	0.94%
14	India	0.65%
15	South Africa	0.60%
16	Argentina	0.52%
17	Saudi Arabia	0.46%
18	Indonesia	0.28%
19	Mexico	0.27%
G20 Average		3.03%

The Indonesian government now plans to raise R&D spending to 1 percent of GDP within a few years [\[31\]](#). Besides public funds, the government is pushing enterprises to increase R&D; for example, it has introduced a tax super-deduction of up to 300 percent to encourage R&D investment in the semiconductor industry [\[32\]](#). How well this works remains to be seen, but given that Indonesia's "state capitalism model" favors industrial policy, it should have some effect.

3. Development of Mid- and High-End Industries

(1) NMC battery plants

Nickel ore is a critical mineral for the battery industry, and Indonesia holds 42 percent of world reserves — an absolutely dominant position. The Indonesian government has therefore used its position as a key mineral supplier to attract world-class enterprises to produce locally.

The integrated battery-industry-chain project in Indonesia (a national strategic project, PSN) [\[33\]](#), a joint venture between an Indonesian state enterprise and China's private company CATL, is already under construction; it is said to be commissioning equipment and will come into production within 2026. Total investment is nearly \$6 billion, with the Chinese firm holding 60 percent and the

Indonesian state enterprise 40 percent. The new battery plant is expected to have an annual capacity of 15 GWh, enough to power 250,000–300,000 EVs.

Besides CATL, the Indonesian government, together with China's EVE Energy and Huayou Cobalt, is preparing to launch another full-chain battery project. Its total scale also reaches \$6 billion ^[34], with planned cell capacity of 20 GWh ^[35].

Both projects cover the entire chain, including cell production. By comparison, India's cell capacity is only 2 GWh, and its efforts to import production lines from China are presently blocked for geopolitical and other reasons. These two projects are thus of great significance to Indonesia; once in production, each will generate billions of dollars of output a year. If they operate well, the government will likely invest more and build additional battery plants.

(2) Semiconductors and photovoltaics

In the semiconductor production chain, Indonesia already has Infineon's packaging-and-testing plant on Batam, employing 2,000 workers. In other segments Indonesia has essentially no capacity. The Ministry of Industry has therefore launched a semiconductor development plan to build chip design and even wafer foundry capability. The government is increasing investment, planning to put in up to \$1 billion ^[36] to build a semiconductor ecosystem. Indonesia's sovereign wealth fund has signed with ARM to jointly train 15,000 chip design engineers ^[37].

In photovoltaics, Indonesia's sovereign wealth fund is leading an international consortium to build a new 50 GW PV module factory. The project has already locked in \$1.4 billion in FDI ^[38].

Like nickel, Indonesia also holds the critical mineral for semiconductors: silica sand. Indonesia's proven silica sand reserves reach 340 million tonnes, more than 15 percent of the world total. The government plans to ban silicon-sand exports entirely and push PV and semiconductor firms to produce locally ^[39]. In the early phase PV firms will mainly land; once Indonesia's semiconductor industry reaches a certain scale, some firms may be willing to cooperate with the government on a wafer foundry.

Indonesia's foundations in semiconductors and PV are both thin; without strong organization by an institution like the sovereign wealth fund, these industries would be very hard to develop.

(3) Shipbuilding

Indonesia is a land of a thousand islands with a vast maritime market and abundant labor, giving it natural advantages in shipbuilding. Its sovereign wealth fund, Danantara, is pushing a major merger and reorganization of as many as 22 state shipbuilding enterprises. In February 2026, Danantara announced the start of a full review of state shipyards' consolidation. The fund has explicitly instructed the naval shipbuilding giant PT PAL Indonesia to act as parent company and reorganize

into a national shipbuilding holding group ^[40]. Danantara is also organizing research institutions and the finance ministry to support shipbuilding from the R&D and financing angles. The aim of these measures is to accelerate localization of shipbuilding ^[41].

In fact, Indonesia's shipbuilding industry already has real strength — not only for domestic use but for export. The largest merchant ship built domestically is the 50,000-tonne Star 50 double-hull bulk carrier ^[42]. The largest warship built in Indonesia, the KRI Balaputradewa (322) missile frigate, displaces more than 6,000 tonnes ^[43]. PT PAL is currently building for the UAE Navy a 163-meter landing platform dock (LPD) of about 13,000 tonnes full load ^[44].

With the full drive of the Indonesian government and rising domestic shipping demand, Indonesia's shipbuilding industry is bound to develop greatly.

4. The Military Industry

Indonesia's military industry is somewhat weaker than Vietnam's, but it is not without foundations; and in the multipolar era, Indonesia has more opportunities to acquire technology from other great powers and middle powers. Building on imported technology, and relying on its own economic strength and human resources, it may develop advanced equipment suited to its own conditions through increased R&D.

Indonesia has signed a technology-transfer agreement with China, which authorizes local firms (such as PT Pindad and PT PAL) to set up assembly and maintenance lines for the C-705 missile ^[45]. Indonesia thus gained anti-ship missile manufacturing capability (localization rate unknown). In 2022, Indonesia launched a reverse-engineering R&D project for this missile, though the results are unknown ^[46]. In a 2026 exercise, the Indonesian navy used C-705 missiles against land targets, possibly showing a deeper mastery of the missile ^[47]. The C-705 has a range of 140 km. With it, the Indonesian military can blockade the Strait of Malacca.

On the army side, the "Harimau" medium tank, co-developed with Turkey and assembled locally in Indonesia, has reached a localization rate of about 50 percent ^[48], and the military plans to equip it on a large scale ^[49]. In artillery, Indonesia lacks the capability to make large-caliber guns, producing only mortars and vehicle-mounted autocannons. Still, through indigenous R&D and technology transfer, it has built up local production of ammunition ^[50] ^[51] ^[52].

In UAVs, Indonesia's homegrown "Elang Hitam" (Black Eagle) surveillance-and-strike drone is said to be fully domestic except for the engine ^[53]. It has a range of 3,000 km, a combat radius of 250 km (without satellite communication, relying on ground stations), and a 24-hour endurance. Indonesia has also developed a kamikaze drone ^[54]; its engine and batteries rely on the civilian supply chain,

while everything else, including the warhead, is locally produced. The maximum strike distance of this kamikaze drone has reached 80 km.

Beyond these indigenous projects, Indonesia is actively importing various missiles and drones from Turkey for local production and assembly.

In an age ruled by missiles and drones, Indonesia's short-term goal is to build deterrence over the Strait of Malacca and the South China Sea shipping lanes; its long-term plan should be to develop UAVs and missiles with ranges above 2,000 km. Once it develops asymmetric capabilities of this range, it can threaten the core industrial areas of the great northern power, reducing the risk of invasion. Yes — China has already risen to be a world power; Indonesia can no longer massacre its Chinese residents, and it only needs to think about not being bullied.



V. Conclusion

Based on analysis of its industrial level, export scale, and class structure, we can confirm that Indonesia is already a capitalist country close to completing industrialization, and should not be regarded as semi-feudal. At the same time, the Indonesian government enjoys complete sovereignty

over import-export tariffs, enterprise taxation, and ownership of key minerals, and should not be regarded as semi-colonial. Indonesia is an industrial country on the "semi-periphery" of the capitalist world-system, now receiving large-scale industrial relocation from the Northern countries and from China. In about fifteen years Indonesia will complete its industrialization and begin developing mid- and high-end industry. To marshal resources and implement industrial policy effectively, Indonesia has gradually taken the "state capitalist road" [55]. It is foreseeable that, like Morocco and Saudi Arabia, Indonesia will use its own resources to develop native industries competitive in the global market, along with a certain level of defense industry.

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